Quality-Adaptive Loss Reweighting Using Expert-Assigned Image Quality for Robust Skin Lesion Classification

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***Abstract*—Diagnostic image quality varies widely in clinical practice, yet quality-aware training methods have had to estimate quality from image statistics or feature norms rather than observe it, because expert judgments of image quality are almost never recorded alongside diagnostic labels. This paper has proposed quality-adaptive loss reweighting, which scales each training sample's classification loss by a real expert-assigned rating of its diagnostic image. Two opposing directions have been evaluated: down-weighting low-quality samples as less reliable, and up-weighting them to force the model to learn from difficult images. The second, the opposite of what the closest face recognition precedents apply, has improved balanced accuracy by 0.037 and malignant sensitivity by 0.092 over an identical baseline. Because a gain of this size is comparable to the fold-to-fold standard deviation, we have validated it causally rather than statistically: a shuffled-quality control, preserving the weight distribution but permuting the lesion-to-rating assignment, has removed the gain entirely and returned performance to baseline, establishing that the improvement has depended on the quality signal's information content rather than on generic reweighting. The full pipeline has reached 0.840 balanced accuracy, of which 65% of the total gain has been attributable to the quality-adaptive loss and the remainder to generic evaluation-time averaging. The method has been made possible by MCR-SL, the only public skin lesion dataset recording per-image expert quality ratings, on which we have also reported the first published benchmark and six robustness analyses, three of which have reversed an apparent positive result.**

***Index Terms*—skin lesion classification, multimodal fusion, dermoscopy, image quality, metadata fusion, deep learning, medical imaging benchmark**

I. Introduction

Medical images acquired in routine practice vary substantially in diagnostic quality. Lighting, focus, framing, and hair occlusion all degrade the evidence a classifier receives, and training as though every image were equally informative assumes a uniformity clinical data does not have. Face recognition has addressed this by making the objective quality aware: AdaFace [14] and MagFace [15] adapt each sample's margin from an estimate of image quality derived from the feature norm, both reducing emphasis on difficult low-quality samples. That reliance on an estimated proxy is a necessity, not a preference, because face datasets do not record what an expert thought of each photograph.

Skin lesion classification has followed a parallel path. Convolutional networks have reached dermatologist-level accuracy on large single-modality archives [2], [3], and a smaller line of work has fused images with patient metadata [4], [5]. In neither case has a per-image expert judgment of quality been available, so quality has remained something to be inferred rather than observed.

MCR-SL [1] has removed that constraint, and is the reason the method proposed here is possible at all. It has paired 779 clinical and 1352 dermoscopic images across 240 lesions from 60 subjects with twenty-two subject-level metadata fields and, distinctively, with a one-to-ten quality rating that dermatologists have assigned to the specific image used for each lesion's diagnosis. It is, to our knowledge, the only public skin lesion dataset carrying such a field, which makes it the enabling dataset for a loss function driven by real expert quality rather than an estimated proxy, not merely a convenient testbed for a pre-existing method.

This paper has therefore asked two separable questions: whether performance degrades on lower-quality images, and whether the training objective can use the quality signal to improve robustness. The answers have differed, the first a null result and the second a positive one validated by a causal control.

A. Contributions

This paper has made the following contributions.

- **Quality-adaptive loss reweighting**, a scheme that scales each sample's classification loss by a real expert-assigned image-quality rating. The direction that has worked, up-weighting low-quality samples, is the opposite of the face recognition precedents that motivated it, and we have offered a domain-specific explanation for why.
- **A shuffled-quality control validating that mechanism causally.** Permuting the lesion-to-rating assignment while preserving the weight distribution has removed the entire gain, establishing that it has depended on the quality signal's information content and not on generic per-sample reweighting.

- A first published benchmark on MCR-SL under a subject-disjoint stratified five-fold protocol, together with a sweep of eleven follow-up configurations that has isolated which interventions help at this scale and which do not.
- Six robustness analyses enabled by MCR-SL's quality ratings and partial histopathology confirmation, including three cases in which an apparent positive finding has not survived a control and has been reported as reversed.

II. Literature Review

Image-only dermoscopic classification has been studied extensively since Esteva *et al.* [2] matched dermatologist performance on a curated archive, with HAM10000 [3] becoming a standard benchmark. These studies have treated the image as the sole input.

Metadata fusion is a distinct thread. Pacheco and Krohling [4] have proposed the Metadata Processing Block, which reweights image features using patient metadata, evaluated on PAD-UFES-20 [5] (2298 clinical images, 1641 lesions). That dataset is the closest comparator in problem shape and is our primary external reference point (Section IV-I).

Quality-aware training has been developed most fully in face recognition. AdaFace [14] adapts the angular margin using the feature norm as a quality proxy, reducing emphasis on hard examples when an image is estimated to be low quality, since a degraded and difficult face is likely unidentifiable and forcing separation would fit noise. MagFace [15] ties feature magnitude to quality similarly. Both depend on an estimated proxy rather than a recorded human judgment.

Within medical imaging, quality has more often been handled architecturally. Quality-guided dynamic routing for diabetic retinopathy [17] conditions the computation path on input quality; that modifies the architecture, whereas the mechanism proposed here modifies only the per-sample loss weight.

Closest to this work, Aburaed *et al.* [16] have applied expert-informed loss weighting on HAM10000, weighting samples by how their diagnostic label was confirmed. That signal describes the reliability of the *label*; ours describes the quality of the *photograph*, a different quantity answering a different question. Our fusion module draws on Squeeze-and-Excitation [6] and is used as an established component, not claimed as a contribution.

A. Research Gap

To the authors' knowledge, no prior work has used a direct expert rating of image quality as a loss-reweighting signal in medical image classification. Existing quality-aware objectives have relied on estimated proxies such as the feature norm [14], [15], existing medical quality-aware methods have acted on the architecture rather than the loss [17], and the closest expert-informed reweighting scheme has weighted by label-confirmation method rather than by image quality [16]. This paper has addressed that gap directly.

III. Proposed Methodology

Fig. 1 summarizes the architecture. This paper has proposed a channel-gated fusion method for cross-modal skin lesion classification and has compared it against two baselines under an identical training and evaluation protocol.

A. Architecture

Fig. 1 shows the pipeline. An ImageNet-pretrained EfficientNet-B0 [7], [8] encodes the dermoscopic image, returning both a 1280-dimensional pooled vector and the final convolutional feature map. A metadata encoder embeds sixteen categorical fields (twelve dimensions each, with a reserved index for missing values, which are never imputed) and four numeric fields (z-scored on training-fold statistics only, each paired with a missingness indicator), and passes their concatenation through a two-layer perceptron to a 128-dimensional vector.

Two fusion variants share a common 256-dimensional projection. The late-fusion baseline concatenates the pooled image and metadata vectors. The channel-gated variant maps the metadata vector through a linear layer and sigmoid to a 1280-dimensional gate, multiplies the feature map channel-wise by that gate, and then pools. This is the Squeeze-and-Excitation mechanism [6] conditioned on metadata rather than on the feature map itself, and is used here as an established component rather than claimed as a contribution of this paper. A binary head and a nine-class auxiliary head sit on the fused vector; a third head predicting the quality rating is added only for the quality-aware variant reported as a negative result below.

B. Training and Evaluation Protocol

Every configuration has been evaluated under subject-disjoint stratified five-fold cross-validation, assigning each subject's lesions to exactly one fold so that no subject or near-duplicate lesion image has appeared in both a training and an evaluation fold. Folds have been balanced by greedily assigning subjects to minimize the difference in malignant lesion count across folds. For each of the five rotations, one fold has been held out as the reported test fold and a second, disjoint fold has been held out purely for checkpoint selection by validation balanced accuracy; the test fold has never influenced model selection, and no hyperparameter has been tuned against final test-fold results.

Seven follow-up interventions have additionally been evaluated on the channel-gated method under the identical protocol: focal loss [12], dermoscopy preprocessing (hair removal and gray-world color normalization), a supervised contrastive auxiliary loss [11], an alternative optimizer using decoupled weight decay with a cosine schedule [10], sharpness-aware minimization [9], test-time augmentation by horizontal flip averaging, and averaging predictions across every dermoscopic image of a lesion. The last two are evaluation-time only. Four stacked combinations bring the total to eleven configurations.

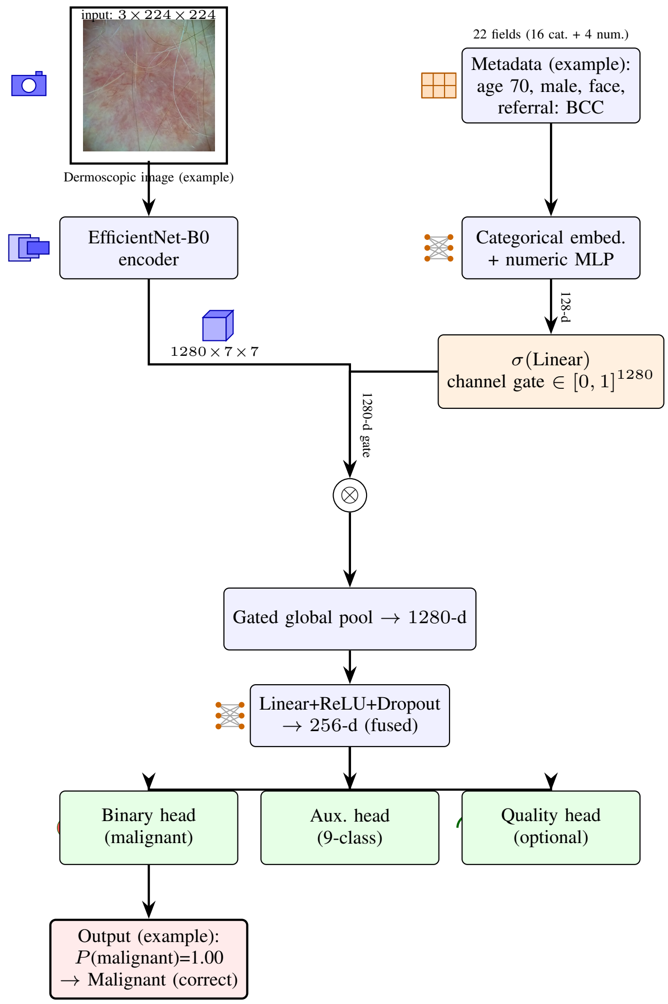

Fig. 1. Channel-gated fusion architecture, with tensor shapes and one real example flowing through: lesion L0189, a histopathology-confirmed basal cell carcinoma, one of the model's highest-confidence correct predictions ($\otimes$: elementwise multiplication of the metadata-derived gate with the convolutional feature map, per channel, before pooling). The late-fusion baseline shares the same final Linear+ReLU+Dropout projection, applied instead to concat(pooled image, metadata) directly.

C. Quality-Adaptive Loss Reweighting

The paper's methodological contribution modifies neither the architecture nor the class of loss function, but the weight each sample carries within the existing objective. Let $q_i \in [1, 10]$ be the mean expert quality rating of lesion i's diagnostic image. A per-sample multiplicative weight w_i^{qual} has been applied alongside the existing per-fold class weight w_i^{cls}:

$$\mathcal{L}_i = w_i^{\text{cls}} \cdot w_i^{\text{qual}} \cdot \text{BCE}(\hat{y}_i, y_i). \tag{1}$$

Two opposing directions have been evaluated, each mapping the rating range onto $[0.5, 1.5]$:

$$\textit{trust}: \quad w_i^{\text{qual}} = 0.5 + (q_i - 1)/9, \tag{2}$$

$$\textit{hard\_mining}: \quad w_i^{\text{qual}} = 1.5 - (q_i - 1)/9. \tag{3}$$

The *trust* direction has treated low-quality images as less reliable and down-weighted them. The *hard_mining* direction has done the opposite, forcing the model to work harder on degraded images. Lesions without a valid rating have received a neutral weight of 1.0. The range has been fixed rather than searched, and only these two directions have been evaluated, so that the comparison isolates direction rather than magnitude.

Only *hard_mining* has survived validation (Sections IV-D–IV-E), and its direction is the opposite of AdaFace's and MagFace's [14], [15]. A plausible domain-specific explanation is that the two settings degrade differently. In face identification a sufficiently degraded photograph can make the identity unrecoverable, so emphasizing such a sample risks fitting noise. In dermoscopy the diagnostic label was in fact assigned from that same image, so the label remains valid and the sample remains an informative hard example.

This mechanism is distinct from the auxiliary quality-regression head described in Section III-A, which has instead asked the model to *predict* the rating as a secondary output. That variant adds a competing objective to a shared representation, whereas reweighting adds no parameters and changes only which samples dominate the gradient. The two are mechanistically different interventions, and the auxiliary head's negative result is reported separately and remains valid; it has not been superseded by the reweighting result.

IV. Experimental Results

A. Dataset

MCR-SL [1] comprises 240 lesions from 60 subjects, 42 malignant and 192 non-malignant, with 1352 dermoscopic and 779 clinical images, 28 lesions histopathology-confirmed, and 16 categorical plus 4 numeric metadata fields. Six lesions carry an undetermined malignancy label and are excluded from the binary task, leaving 234 usable. Twenty-one lesion identifiers in the image index have no lesion-table row and no file on disk, and are dropped via an inner join. Section IV-G traces the further reduction to the 231 lesions used in the analyses.

B. Evaluation Metrics

We have reported accuracy, balanced accuracy, macro-averaged F1, sensitivity on the malignant class, specificity, and area under the receiver operating characteristic curve, AUROC, at every shot of evaluation, aggregated across the five test folds as mean and standard deviation. Balanced accuracy has been used for checkpoint selection and as the primary comparison metric, since the malignant class has represented a minority of usable lesions.

C. Core Ablation Results

Table I reports the core ablation at three-decimal precision, since several values are too close for two decimals to separate. Channel-gated fusion has reached the highest macro-F1 but has not dominated: the image-only baseline has higher sensitivity (0.697 vs. 0.672) and AUROC (0.895 vs. 0.881), and its balanced accuracy is effectively tied (0.784 vs. 0.783, against a standard deviation of 0.07–0.09). We report this rather than framing it as a clean win for the fusion method.

The quality-aware variant, which adds an auxiliary head predicting the quality rating, has underperformed on five of six metrics: balanced accuracy 0.740 vs. 0.783 and sensitivity 0.578 vs. 0.672, against a small specificity gain (0.902 vs. 0.895). It has become more conservative, trading a large sensitivity loss for a small specificity gain, the worse direction for cancer screening. This is a negative result for *predicting* quality, and is mechanistically distinct from the reweighting scheme of Section III-C.

TABLE I
CORE ABLATION MATRIX, MEAN ± STD OVER FIVE FOLDS

Metric	Image-only	Late fusion	Channel-gated
Accuracy	0.813 ± 0.020	0.831 ± 0.052	0.833 ± 0.070
Balanced acc.	0.784 ± 0.073	0.768 ± 0.090	0.783 ± 0.085
Macro F1	0.733 ± 0.038	0.743 ± 0.051	0.757 ± 0.073
Sensitivity	0.697 ± 0.187	0.644 ± 0.191	0.672 ± 0.159
Specificity	0.872 ± 0.061	0.891 ± 0.021	0.895 ± 0.048
AUROC	0.895 ± 0.053	0.851 ± 0.080	0.881 ± 0.065

Channel-gated, quality-aware variant: accuracy 0.812 ± 0.068, balanced accuracy 0.740 ± 0.065, macro F1 0.715 ± 0.065, sensitivity 0.578 ± 0.177, specificity 0.902 ± 0.073, AUROC 0.879 ± 0.041.

D. Quality-Adaptive Loss Results

Both weighting directions of Section III-C have been trained under the identical protocol, so the loss weighting is the only difference between them and the baseline (Table II). The *hard_mining* direction has improved every core metric: balanced accuracy by 0.037 (0.783 to 0.820), malignant sensitivity by 0.092 (0.672 to 0.764), AUROC by 0.025 (0.881 to 0.906), and accuracy from 0.833 to 0.845, against a small specificity cost (0.895 to 0.875). The sensitivity gain is the clinically weightier, since a missed malignancy is costlier than a false alarm. The *trust* direction has been approximately flat on balanced accuracy (0.788) and is not treated as an improvement. Since 0.037 is smaller than the fold-to-fold standard deviation of roughly 0.08, Section IV-E tests it causally rather than resting on its magnitude.

E. Validation via Shuffled-Quality Control

The obvious objection is that any per-sample reweighting might act as a regularizer, in which case the ratings would be incidental. We have tested this directly: within each fold's training portion the ratings have been permuted across lesions under a fixed seed, preserving the weight distribution exactly while destroying the lesion-to-rating correspondence. Every other aspect of training is identical, and evaluation always uses the true ratings, since only the training-time weight computation is affected.

Under the shuffled control, *hard_mining*'s balanced accuracy has fallen from 0.820 to 0.775 and its sensitivity from 0.764 to 0.667, both returning to the baseline's level (0.783 and 0.672). A distribution-matched but uninformative weighting has recovered none of the gain, establishing that the improvement depends on the weights tracking real quality rather than on reweighting as such. This is a causal argument rather than a statistical one, and it is why the result is reported despite an effect size comparable to the fold-to-fold standard deviation.

The control has also been informative negatively. The *trust* variant's one apparent advantage, a narrower accuracy gap between the highest and lowest quality terciles (0.082 against 0.091), has narrowed further still under shuffling, to 0.021: a weighting carrying no quality information has produced a flatter profile than the mechanism intended to produce one. That advantage is therefore reported as a reversed finding rather than a second positive result.

TABLE II
SHUFFLED-QUALITY CONTROL. SHUFFLING PERMUTES THE LESION-TO-RATING ASSIGNMENT WITHIN EACH TRAINING FOLD WHILE PRESERVING THE WEIGHT DISTRIBUTION EXACTLY

Configuration	Bal. acc.	Sens.	High−low gap
Plain baseline (no quality)	0.783	0.672	0.091
hard_mining, real ratings	**0.820**	**0.764**	0.091
hard_mining, shuffled	0.775	0.667	0.016
trust, real ratings	0.788	0.725	0.082
trust, shuffled	0.763	0.672	0.021

F. Follow-Up Intervention Sweep

Eleven follow-up configurations have been evaluated on the channel-gated method. Sharpness-aware minimization with test-time augmentation has reached 0.822 balanced accuracy, the best of this sweep; focal loss has raised specificity to 0.923 and dermoscopy preprocessing has given the sweep's highest sensitivity, 0.736, trading off in opposite directions. Contrastive pretraining, the alternative optimizer, and quality-aware training have each underperformed the baseline, a convergent negative result across three independent interventions. Stacking every promising intervention together has underperformed both the baseline and its components, 0.759 against 0.783 and 0.822, plausibly because focal loss and preprocessing pull the decision boundary in opposite directions.

Applying the two evaluation-time interventions on top of *hard_mining*, rather than on top of the plain baseline, has produced this paper's best configuration. Table III decomposes it. Test-time augmentation has raised balanced accuracy to 0.833 and sensitivity to 0.808, the highest sensitivity of any configuration reported here; adding multi-image averaging has raised balanced accuracy further to 0.840 ± 0.095. Notably, multi-image averaging has stacked cleanly here, whereas it reduced balanced accuracy when combined with sharpness-aware minimization and test-time augmentation in the sweep above; that non-additivity has therefore been specific to that combination rather than a general property of the intervention.

The decomposition matters for how the headline number should be read. Of the total 0.057 gain over the baseline, 0.037, or roughly 65%, has come from the quality-adaptive loss and has been causally validated in Section IV-E. The remaining 0.020 has come from generic evaluation-time averaging that requires no retraining and has nothing to do with image quality. We state this split explicitly so that the contribution attributable to the proposed mechanism is not overstated by the final figure.

TABLE III
DECOMPOSITION OF THE BEST CONFIGURATION. ONLY THE FIRST STEP USES THE QUALITY SIGNAL; THE REMAINDER IS GENERIC EVALUATION-TIME AVERAGING

Step	Quality?	Bal. acc.	Δ	Sens.
Channel-gated baseline	no	0.783	—	0.672
+ *hard_mining* loss	**yes**	0.820	+**0.037**	0.764
+ test-time augmentation	no	0.833	+0.050	**0.808**
+ multi-image averaging	no	**0.840**	+0.057	0.783
hard_mining, shuffled	control	0.775	−0.008	0.667

G. Robustness Analyses

These analyses use the channel-gated method's out-of-fold predictions. Of the 234 binary-usable lesions, 231 have received a prediction; three lacked a usable image in their assigned test fold and were excluded rather than imputed. Each of the 231 also carries a valid quality rating.

Quality-stratified performance. Grouping lesions into terciles by mean expert rating gives accuracies of 0.776, 0.914 and 0.868 (low, mid, high) over uneven tercile sizes of 85, 93 and 53, since the mean of up to three integer ratings takes few distinct values and ties cluster at the boundaries. The pattern is not monotonic and the Spearman correlation between rating and prediction error is -0.093 ($p = 0.158$), so we report no strong quality-performance relationship detected at 231 lesions, neither confirming nor ruling out an effect.

The validated *hard_mining* configuration has not flattened this profile: its high-minus-low accuracy gap has remained 0.091. Its benefit has appeared within terciles instead, most visibly on the lowest-quality one, where malignant sensitivity has risen from 0.667 to 0.810. The mechanism has improved detection on poor images without equalizing accuracy across quality levels, and we have not claimed the latter.

Reversed findings. Three apparently positive results have not survived scrutiny and are reported rather than omitted. First, *trust*'s narrower quality-tercile gap was undercut by the shuffled control producing a narrower gap still (Section IV-E). Second, a pooled correlation between expert diagnostic certainty and prediction error, initially significant at $\rho = -0.175$ ($p = 0.008$, $N = 231$), dissolved once stratified by class ($p = 0.083$ and $p = 0.559$): malignant lesions concentrate at low certainty and the model is weaker on them, so the pooled figure reflected class composition rather than an independent relationship. Third, a proposed change to train on all images per lesion was found on verification to describe what the pipeline already did. At this sample size an uncontrolled first look has repeatedly produced positives that controls dissolve, and reporting this pattern is part of the contribution.

Intra-subject consistency. Of 55 subjects with two or more evaluable lesions, perfectly consistent subjects numbered 27 against a permutation null mean of 29.9 ($p = 0.954$), and none had all lesions misclassified, so errors have not concentrated in particular patients.

Histopathology-confirmed versus panel-consensus. The model has performed worse on the 28 histopathology-confirmed lesions than on the 203 panel-consensus-only ones, 0.750 against 0.867 accuracy, plausibly because histopathology is more often ordered for clinically ambiguous lesions. With 28 lesions this has been treated as qualitative only.

H. Qualitative Analysis: Hard Samples

Grad-CAM has been applied to lesions selected by a systematic rule rather than by inspection, one from each of four categories, preferring malignant lesions; Fig. 2 contrasts a correct and an incorrect hard case. Gradients are taken with respect to the feature map *after* the metadata gate, since that is the representation the decision is computed from, with the metadata branch held fixed so attribution reflects image evidence only.

The pair inverts the expected relationship with quality. Panel (a) is a basal cell carcinoma rated 4.3, washed out and hair-obscured, yet attention has localized the lesion and the malignant call has been made at $P = 1.00$. Panel (b) is a melanoma rated 8.3 on a clean image, where attention has also localized the lesion correctly and the prediction has still been benign at $P = 0.20$. Attention is correct in both, so the failure in (b) is one of decision, not of attention, and is not attributable to image quality. This is the null of Section IV-G in visual form; the residual difficulty in (b) is more plausibly melanoma's scarcity, the rarest malignant class here, than any property of the photograph.

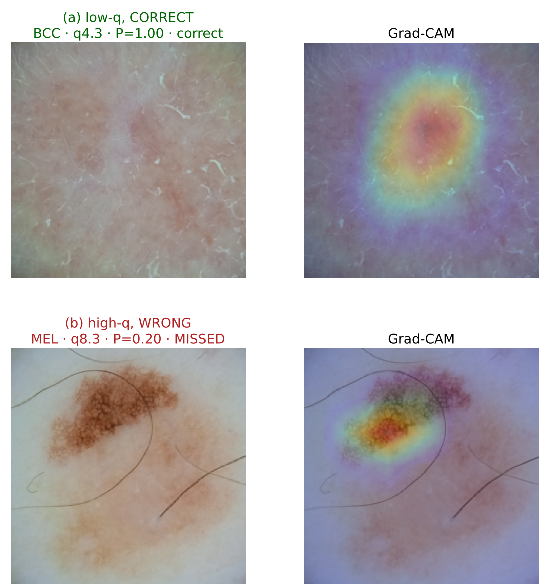

Fig. 2. Grad-CAM on the post-gate feature map. P is $P(\mathrm{malignant})$, q the expert quality rating. (a) basal cell carcinoma correctly identified despite a low-quality, hair-obscured image; (b) melanoma missed on a high-quality image with attention correctly localized. Maps are 7×7 upsampled, indicating approximate attention only.

I. Comparison with the Literature

This paper's channel-gated baseline, *hard_mining* variant, and best configuration have reached 0.783, 0.820 and 0.840 balanced accuracy respectively, against the closest published binary-task work. PAD-UFES-20's six-class balanced accuracies (0.765–0.842) are not valid comparators for a binary task despite the shared metric name. The binary result we have verified directly is Uliana and Krohling's [13], 0.836 from a diffusion classifier over 2298 images under five-fold cross-validation, at the image level rather than our lesion level. This paper's best configuration has reached 0.840 ± 0.095. The nominal 0.004 difference is more than twenty times smaller than our own fold-to-fold standard deviation, so the two results are statistically indistinguishable and we have described ours as within noise of that benchmark rather than as exceeding it. The comparison is also not like for like: the two counts differ in unit, our 234 lesions against their 2298 images, and the underlying datasets differ by roughly sevenfold in lesion count and, more consequentially, in class balance, 17.9% malignant here against 47.5% there. HAM10000 [3] headline results have reached higher raw accuracy, but on a dermoscopy-only task with no metadata and roughly forty times more training images, and have not been treated as comparable.

MCR-SL has had no prior published benchmark, so this result is the current best on it by default; we have avoided framing that as a state-of-the-art claim, since there has been no prior number to surpass, and have emphasized the loss mechanism and its validation (Sections IV-D–IV-E) as the principal contribution.

V. Conclusion and Future Work

This paper has proposed quality-adaptive loss reweighting, scaling each training sample's classification loss by a real expert-assigned rating of its diagnostic image. Up-weighting low-quality samples has improved balanced accuracy by 0.037, sensitivity by 0.092 and AUROC by 0.025 over an identical baseline, and evaluation-time averaging on top has reached 0.840 ± 0.095; of that 0.057 total gain, 65% is attributable to the loss and the remainder to generic technique. The validated direction is the opposite of the face recognition precedents that motivated it [14], [15], plausibly because a dermatologist can usually still diagnose from a degraded photograph while a degraded face can lose its identity entirely.

The effect size deserves care. A gain of 0.037 is comparable to, and in places smaller than, the fold-to-fold standard deviation of roughly 0.08, so this result does not rest on statistical power, which 234 lesions cannot supply. It rests instead on the shuffled-quality control: permuting the lesion-to-rating assignment while preserving the weight distribution has removed the gain entirely, which is causal evidence that the mechanism uses the quality signal's information content. We regard that as the appropriate standard of evidence at this scale, and have reported the effect size plainly rather than relying on the final figure to carry the claim.

Two questions have been answered differently. Performance has shown no significant relationship with expert-rated quality at this sample size, a null that also held for expert certainty once class composition was controlled; yet the training objective has still exploited the quality signal. These are separable questions, not a contradiction.

This study has been limited by MCR-SL's size, 234 usable lesions from one institution with 28 histopathology-confirmed, by a single random seed, and by a 17.9% malignant rate leaving eight or nine malignant lesions per test fold, so one lesion changing outcome moves that fold's sensitivity by about 0.11. Future work should validate the mechanism on other quality-labeled medical datasets, most directly DeepDRiD [18], [19], which records per-image quality alongside diabetic retinopathy grades and which we have prepared but not completed within this cycle, and should give numeric metadata fields representational capacity comparable to categorical embeddings before re-running the importance audit.

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